Digital Payment Fraud Risk Analytics

Detecting High-Value Transaction Leakage Using SQL

Executive Summary

The This project presents a simulated business case for **PaySecure**, a fictional digital payment platform operating in the financial services industry. The analysis is performed using the **PaySim** dataset, which simulates real-world mobile money transactions. As transaction volumes continue to grow, the organization faces increasing financial losses due to fraudulent activities. While fraud represents only a small fraction of overall transactions, its financial impact is significant, making fraud prevention a critical business priority.

This project analyses 6.36 million digital payment transactions from the PaySim dataset to quantify fraud-related losses, identify high-risk transaction patterns, evaluate the effectiveness of the existing fraud detection system, and provide actionable recommendations to strengthen fraud prevention strategies.

The analysis found that fraudulent transactions account for only 0.13% of all transactions but resulted in approximately 12.06 million in financial losses. Fraud is concentrated entirely within TRANSFER and CASH_OUT transactions, while the existing fraud detection mechanism captures only a small portion of fraudulent activity. The findings provide management with data-driven recommendations for reducing financial risk and improving transaction security.

1. Business Context

Company Background

PaySecure is a fictional digital payments company operating within the Financial Services (FinTech) industry. The company provides customers with secure online money transfers and digital payment services while processing millions of financial transactions every day.

This business scenario is analysed using the **PaySim** dataset, which simulates mobile money transactions based on real transaction patterns. The dataset represents the transaction environment used throughout this project.

As the company's transaction volume has increased, fraudulent activities have also become more sophisticated. Fraudsters exploit weaknesses in payment channels to perform unauthorized transfers, resulting in direct financial losses and increased operational risk.

To strengthen fraud prevention efforts, the Risk Management and Fraud Prevention teams initiated a data-driven analysis to better understand fraud behaviour and improve the effectiveness of existing fraud controls.

This project analyses transaction-level data to identify fraud patterns, quantify financial losses, evaluate fraud detection performance, and generate data-driven recommendations that support stronger fraud prevention strategies and business decision-making.

2. Business Problem

Although the simulated mobile money platform includes an existing fraud detection system, management lacks sufficient visibility into fraud performance and financial exposure.

The organization needs answers to several critical business questions.

• How much money is being lost due to fraudulent transactions?

• Which transaction types are most vulnerable to fraud?

• Are fraudulent transactions concentrated during specific periods?

• What characteristics distinguish fraudulent transactions from legitimate ones?

• How effective is the current fraud detection system?

• Which business areas should receive higher fraud monitoring priority?

Without these insights, fraud prevention resources cannot be effectively allocated, allowing fraudulent activities to continue generating financial losses.

3. Business Objective

The objective of this project is to analyse over 6.36 million digital payment transactions using SQL to identify fraud patterns, measure the financial impact of fraudulent activity, evaluate fraud detection effectiveness, and generate actionable recommendations that strengthen fraud prevention strategies and support data-driven business decision-making.

4. Dataset Overview

The analysis is performed on the PaySim financial transactions dataset containing more than 6.36 million digital payment records.

Each transaction includes:

- Transaction Time (Step)
- Transaction Type
- Transaction Amount
- Sender Account
- Receiver Account
- Sender Balance
- Receiver Balance
- Fraud Indicator (isFraud)
- Fraud Detection Flag (isFlaggedFraud)

5. Key Performance Indicators

KPI	Value
Total Transactions	6.36 Million
Fraud Transactions	8,213
Fraud Rate	0.13%
Total Fraud Loss	12,056.42 Million

6. Key Findings

Finding 1 — Fraud Represents a Small Percentage of Transactions but a Significant Financial Risk

Only 8,213 out of 6.36 million transactions were identified as fraudulent, representing approximately 0.13% of total transaction volume.

Despite the low fraud rate, these transactions generated over 12.06 million in financial losses, demonstrating that even a small number of fraudulent activities can have a substantial business impact.

Finding 2 — Fraud is Concentrated Within Specific Transaction Types

Fraud occurred exclusively within TRANSFER and CASH_OUT transactions.

TRANSFER transactions recorded the highest fraud rate, while CASH_OUT transactions generated nearly the same number of fraud cases and financial losses.

No fraudulent activity was observed in PAYMENT, CASH_IN, or DEBIT transactions, indicating that fraud is concentrated within only two transaction channels.

Finding 3 — Fraud Occurs Continuously Throughout the Day

Fraud activity remained relatively consistent across all transaction hours.

There were no significant periods where fraudulent transactions disappeared or sharply increased, suggesting that fraudsters operate continuously rather than targeting specific time windows.

This indicates that fraud monitoring should remain active throughout the entire day.

Finding 4 — Current Fraud Detection Controls Require Improvement

A comparison between actual fraudulent transactions and system-flagged fraud indicates that the existing fraud detection process identifies only a small proportion of fraudulent activity.

This leaves the organization exposed to substantial financial losses and highlights opportunities to improve fraud detection accuracy through stronger monitoring rules and risk scoring techniques.

Finding 5 — Fraudulent Transactions Display Distinct Behavioural Characteristics

Fraudulent transactions typically involve significantly larger transaction amounts compared to legitimate transactions.

They also result in major reductions in sender account balances, with almost all fraudulent transactions leaving the originating account with little or no remaining balance.

These behavioural characteristics provide strong indicators that can be incorporated into future fraud detection models.

Finding 6 — High-Value Transactions Present the Greatest Fraud Risk

High-value transactions account for the majority of fraudulent activity and record the highest fraud rates.

Medium-value transactions contribute fewer fraud cases, while low-value transactions rarely exhibit fraudulent behaviour.

This demonstrates a strong relationship between transaction value and fraud risk.

7. Business Recommendations

Based on the findings, the following actions are recommended to strengthen fraud prevention and reduce financial risk.

- Prioritize fraud monitoring for TRANSFER and CASH_OUT transactions, as these channels account for all identified fraudulent activity.
- Implement additional verification procedures for high-value transactions before payment authorization.
- Enhance fraud detection rules by incorporating sender balance depletion, unusually large transaction amounts, and zero-balance behaviour.
- Recalibrate the fraud detection system to improve recall and increase the proportion of fraudulent transactions successfully detected.
- Deploy continuous real-time fraud monitoring across all operating hours instead of relying on time-specific monitoring.
- Develop a transaction risk scoring model combining transaction type, transaction amount, account balance changes, and customer behaviour to identify suspicious transactions before financial losses occur.

8. Expected Business Impact

Implementing the recommended fraud prevention strategies is expected to deliver several business benefits.

- Reduce financial losses caused by fraudulent transactions.
- Improve the accuracy and effectiveness of fraud detection systems.
- Enable earlier identification of high-risk transactions before payment completion.
- Prioritize fraud prevention resources toward the highest-risk transaction channels.
- Strengthen customer trust through improved transaction security.

- Support better risk management decisions using data-driven fraud insights.
- Improve operational efficiency by focusing investigations on the most suspicious transactions.

9. Conclusion

This project demonstrates how SQL can be used to transform large-scale financial transaction data into meaningful business insights for fraud prevention.

Through the analysis of more than 6.36 million payment transactions, the project quantified fraud-related financial losses, identified high-risk transaction channels, uncovered behavioural characteristics associated with fraud, and evaluated the effectiveness of the existing fraud detection process.

The findings provide a strong analytical foundation for strengthening fraud prevention strategies, reducing financial losses, and supporting data-driven decision-making within financial services organizations.

10. Skills Demonstrated

- SQL Data Cleaning
- Exploratory Data Analysis (EDA)
- Fraud Analytics
- Financial Risk Analysis
- Business KPI Development
- Aggregate Functions
- CASE Statements
- Common Table Expressions (CTEs)
- SQL Views
- Business Reporting
- Executive Dashboard Design
- Data Visualization
- Data Storytelling